MR-SLAM: Immersive Spatial Supervision for Multi-Robot Mapping via Mixed Reality

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***Abstract*—Operating a multi-robot fleet for simultaneous localization and mapping (SLAM) in applications such as building inspection or warehouse-aisle monitoring requires the operator to maintain spatial awareness of each robot's position and mapping state, a task that scales poorly on conventional 2D interfaces. We present MR-SLAM, a mixed reality (MR) system in which an operator wearing a Meta Quest 3 headset teleoperates three simulated TurtleBot3 robots through a passthrough view with real-world occlusion, while spatially anchored dashboard panels report mapping progress in situ. Each robot runs an independent SLAM Toolbox instance whose occupancy grid is merged in real time on a Robot Operating System 2 (ROS 2) back-end. Across five 9-minute evaluation sessions, the system delivered scans at $8.83 \pm 0.16\,$Hz, mapped $17.9 \pm 0.8\,\mathrm{m}^2$ of merged occupancy, and reached $94.7 \pm 0.5\%$ cross-instance occupancy consistency across robot pairs. An additional session recorded 6.3 ms median transform jitter and $26.7\,\mathrm{m}^2$ coverage of a $41\,\mathrm{m}^2$ grid. We position MR-SLAM as a reference implementation for combining passthrough mixed reality supervision with multi-robot SLAM on consumer hardware.**

***Index Terms*—SLAM, Multi-robot Systems, Mixed Reality, Spatial AI, Teleoperation, Human-Robot Interaction, Mobile Robots**

I. Introduction

Simultaneous localization and mapping (SLAM) allows mobile robots to operate without external positioning infrastructure [1], [2]. The problem was first formulated as a joint state estimation task [3] and later grounded in probabilistic robotics [4]. Extending SLAM to multi-robot teams enables faster area coverage and fault tolerance, but introduces challenges in communication, map fusion, and operator supervision [5]. These challenges are seen in time-critical applications such as building inspection and warehouse inventory, where an on-site operator must coordinate multiple robots from a single station. The SLAM back-end has matured from graph-based optimizers [6] to distributed systems [7], [8], yet the operator interface for multi-robot supervision during mapping has received limited attention.

Robot teleoperation typically relies on desktop workstations with 2D displays and joystick or keyboard input [9], [10]. These interfaces require the operator to mentally reconstruct 3D spatial relationships from flat views, and this cognitive burden grows with the number of robots [11]. Virtual Reality (VR), augmented reality (AR), and mixed reality (MR) head-mounted displays (HMDs) have been shown to improve situational awareness in single-robot teleoperation [12]–[14] as well as facilitate safety assessment of autonomous systems [15]. Walker et al. [16] reported a 28% improvement in operator effectiveness using an MR interface for single-robot navigation. However, none addresses the coordination challenges of supervising multiple robots during active SLAM.

In prior work, we developed SimNav-XR [17], an extended reality platform for single-robot simulation integrating Unity, ROS 2, and the Meta Quest 3. SimNav-XR demonstrated MR-based robot control with LiDAR simulation and Nav2 integration but was limited to a single robot without real-time SLAM.

This paper presents MR-SLAM, which extends SimNav-XR to multi-robot SLAM with three contributions: (1) a passthrough MR interface on the Meta Quest 3 rendering three simulated TurtleBot3 robots with raycasted LiDAR, spatial occlusion, and joystick teleoperation; (2) a multi-robot SLAM pipeline with independent SLAM Toolbox [18] instances and real-time map merging via `multirobot_map_merge` [19], [20]; and (3) a spatially anchored dashboard design displaying coverage, overlap, scan rate, and latency metrics. The system targets scenarios where an operator supervises a small robot fleet from a fixed location, such as inspecting a building wing or monitoring warehouse aisles.

II. Related Work

A. Multi-Robot SLAM

Multi-robot SLAM architectures range from centralized to fully decentralized [5], [21]. Centralized systems such as CCM-SLAM [22] aggregate data at a server for joint optimization; this simplifies map fusion but introduces a communication bottleneck and a single point of failure. Distributed systems such as DOOR-SLAM [7] and Kimera-Multi [23], [24] perform peer-to-peer pose-graph optimization with outlier rejection, improving scalability at the cost of increased synchronization complexity. Swarm-SLAM [8] provided a ROS 2 decentralized framework supporting multiple sensor modalities, and LAMP [25] and LAMP 2.0 [26] addressed

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subterranean multi-robot SLAM during the DARPA SubT Challenge.

Occupancy grid mapping [27], [28] remains standard for 2D LiDAR. ORB-SLAM2 [29] established feature-based visual SLAM, and Google Cartographer [30] demonstrated real-time 2D LiDAR SLAM with branch-and-bound loop closure. For ROS 2 specifically, Macenski et al. developed SLAM Toolbox [18] as a lifelong 2D SLAM solution, and benchmarked modern visual SLAM approaches for Nav2 integration [31]. Map merging for occupancy grids was formalized by Carpin and Birk [19], [32] using spectral methods and stochastic search, and extended with Hough-transform acceleration [20]. The `multirobot_map_merge` ROS package has been applied in multi-TurtleBot3 deployments [33], [34].

B. Mixed Reality for Robot Teleoperation

Surveys by Suzuki et al. [12] and Walker et al. [13] catalogued AR/MR strategies for human-robot interaction (HRI). Early work demonstrated intent communication via HMD overlays [14], [35]. Whitney et al. [36] introduced ROS Reality for Unity-based virtual reality (VR) teleoperation. On the Meta Quest platform, Quest2ROS [37] measured 0.46 mm tracking accuracy with 82 ms latency, and Quest2ROS2 [38] extended this to ROS 2. Morando and Loianno [39] demonstrated MR-based human-drone collaboration with spatial awareness sharing. Roldán et al. [40] addressed multi-robot VR monitoring without SLAM, and the Cyber-Physical Control Room [16] provided MR perspectives for single-robot teleoperation. However, none addresses the cognitive and coordination challenges of supervising multiple robots during active SLAM.

C. ROS–Unity Integration

The Unity Robotics Hub [41] provides `ros_tcp_connector` for Unity–ROS communication. Allspaw et al. [42] benchmarked Unity–ROS bridges and found `ros_tcp_connector` offers the best latency-integration trade-off. Platt and Ricks [43] showed that Unity matches or exceeds Gazebo for SLAM simulation with better scalability. These integration frameworks enable real-time bidirectional communication, yet none has been applied to multi-robot SLAM with an immersive operator interface.

D. Positioning of This Work

Table I compares MR-SLAM with related systems. Prior work addresses either MR-based single-robot teleoperation [16], [37], [39] or multi-robot SLAM without an MR interface [8], [33]. Roldán et al. [40] built a multi-robot VR interface but did not integrate SLAM. Patel et al. [44] explored MR for multi-robot interaction but without a SLAM pipeline. In contrast, MR-SLAM uniquely combines passthrough MR visualization, real-time multi-robot SLAM with occupancy grid merging, and spatial performance feedback into a unified system, enabling operators to supervise mapping tasks directly within their physical environment.

TABLE I
COMPARISON WITH RELATED SYSTEMS

System	MR Pass.	Multi Robot	SLAM	Dash.
Cyber-Phys. CR [16]	✓	–	–	✓
Quest2ROS [37]	–	–	–	–
Morando et al. [39]	✓	–	–	–
Roldán et al. [40]	–	✓	–	✓
Patel et al. [44]	✓	✓	–	–
Swarm-SLAM [8]	–	✓	✓	–
Nica et al. [33]	–	✓	✓	–
SimNav-XR [17]	✓	–	✓	–
MR-SLAM (ours)	✓	✓	✓	✓

III. System Architecture

MR-SLAM consists of a Unity-based MR application on the Meta Quest 3 and a ROS 2 SLAM back-end on an Ubuntu 22.04 laptop, connected over WiFi. All robots are fully simulated within Unity; no physical robots are used. The operator wears the headset in a real room, sees virtual robots overlaid on the physical environment through passthrough rendering, and issues velocity commands via the controller. These commands drive simulated differential-drive physics, generating LiDAR scans and transform data streamed to ROS 2 for SLAM processing; resulting maps and statistics are sent back for dashboard display. Fig. 1 shows the layered architecture.

A. MR Application Layer

The application runs in Unity 2022 LTS on Android (Meta Quest 3). Meta's Mixed Reality Utility Kit (MRUK) provides room-scale passthrough and spatial occlusion, using on-device AI to reconstruct a spatial mesh of the physical environment from depth and color data: virtual robots behind real furniture are correctly hidden (Fig. 2). Three TurtleBot3 Burger models [45], [46] use differential-drive physics controllers with articulated wheel joints.

Each robot carries a simulated 2D LiDAR implemented via Unity raycasting—180 rays over a 90° forward-facing arc at 10 Hz—producing `LaserScan` messages on namespaced topics (`/robot1/scan`, etc.). This configuration approximates the forward field of view of a planar LiDAR while maintaining real-time performance on the headset's mobile processor. A TF publisher broadcasts `odom`→`base_footprint` and child frames at 20 Hz. A clock publisher sends simulation time at 100 Hz via `/clock`. Teleoperation uses the Meta Quest 3 right Touch controller thumbstick to issue `Twist` commands, clamped to 0.15 m/s linear and 1.5 rad/s angular velocity.

B. Spatial Dashboard

World-space Canvas panels anchored in the MR environment provide real-time fleet telemetry (Fig. 3). A fleet overview panel shows each robot's connection status and scan rate. Per-robot panels display map dimensions, coverage

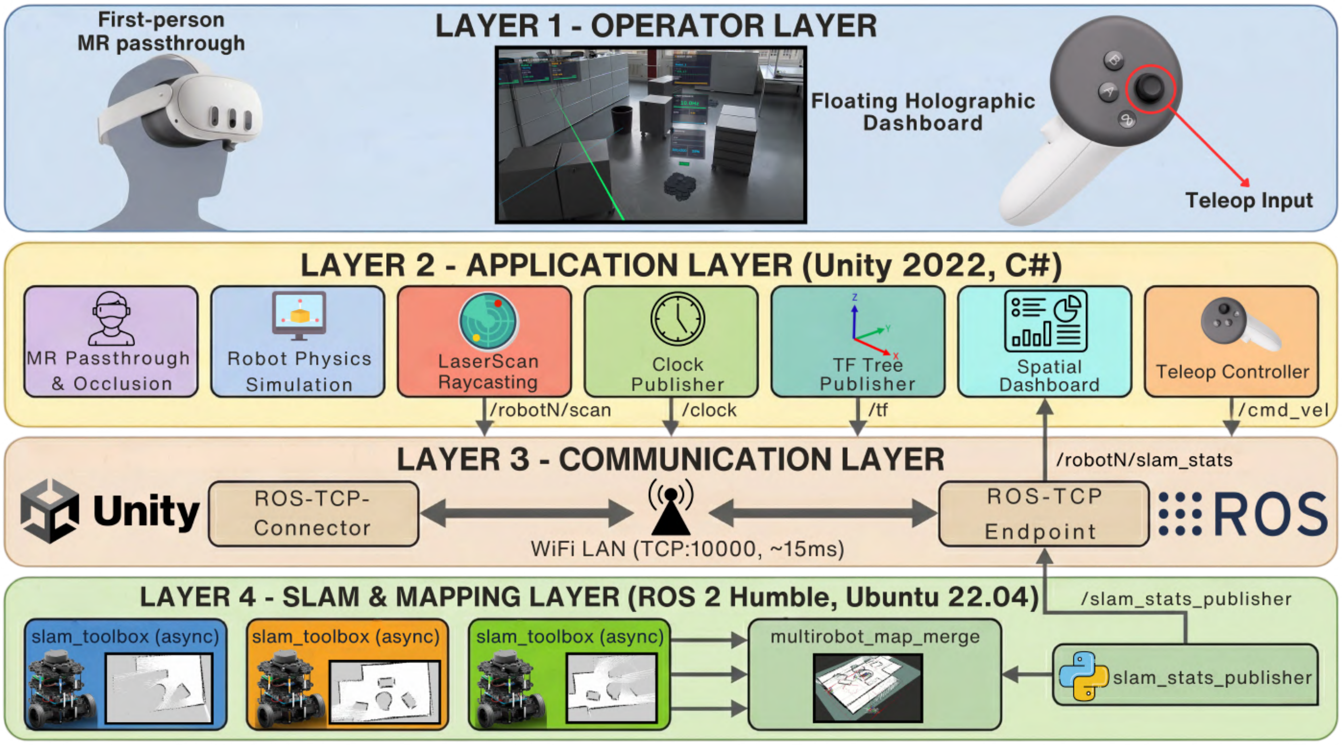

Fig. 1. MR-SLAM system architecture. Layer 1: operator with Meta Quest 3 passthrough MR. Layer 2: Unity application with robot physics, LiDAR raycasting, TF publishing, spatial dashboard, and teleop controller. Layer 3: `ros_tcp_connector` bridge over WiFi (TCP:10000). Layer 4: three independent SLAM Toolbox instances produce per-robot maps that are fused by `multirobot_map_merge`.

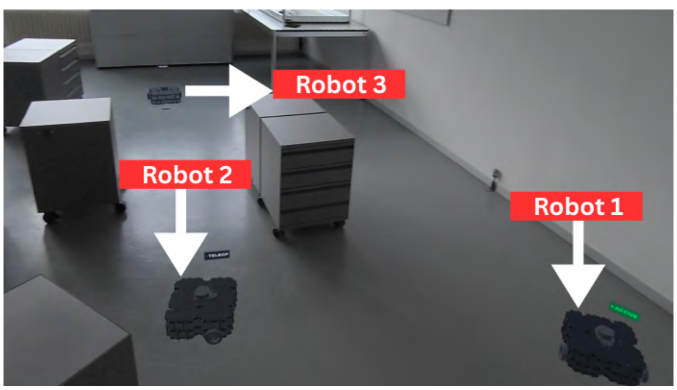

Fig. 2. Three simulated TurtleBot3 robots placed in the laboratory environment as seen through Meta Quest 3 passthrough. MRUK spatial occlusion hides robots behind real-world furniture.

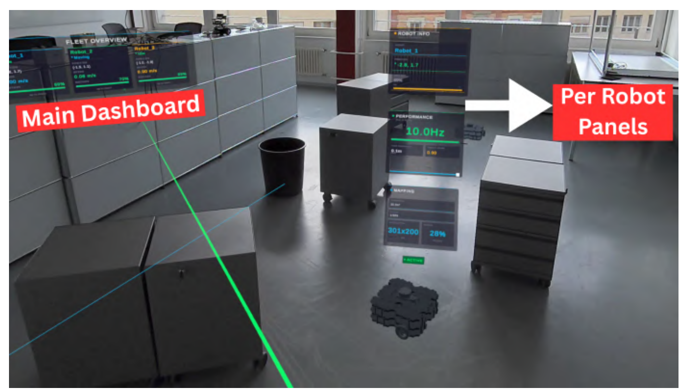

Fig. 3. In-headset spatial dashboard viewed through passthrough MR. Left: fleet overview. Right: per-robot panels showing scan rate (10.0 Hz), map dimensions, and coverage.

area, and the latest scan timestamp, updated at 1 Hz from the `slam_stats_publisher` node.

C. Communication Bridge

Bidirectional communication uses `ros_tcp_connector` [41] (Unity side) and `ros_tcp_endpoint` (ROS 2 side) over TCP port 10000. Table II lists the primary topics.

TABLE II
PRIMARY ROS 2 TOPICS

Topic	Type	Rate	Dir.
`/robotN/scan`	LaserScan	10 Hz	U→R
`/tf`	TFMessage	20 Hz	U→R
`/clock`	Clock	100 Hz	U→R
`/cmd_vel[_N]`	Twist	on-demand	U→R
`/robotN/map`	OccupancyGrid	∼1 Hz	R→U
`/robotN/slam_stats`	Float32MultiArray	1 Hz	R→U

U = Unity (Meta Quest 3), R = ROS 2. N $\in \{1, 2, 3\}$.

D. SLAM and Map Merging

On the ROS 2 back-end (Intel i5, 16 GB RAM), three SLAM Toolbox [18] instances run in asynchronous mode, each subscribing to a namespaced scan topic. Asynchronous mode was selected for its tolerance of variable-latency scan delivery over WiFi and its support for lifelong mapping with serialization. The `multirobot_map_merge` node subscribes to all three maps and produces a merged occupancy grid. Because the robots share a common `map` frame, the merge uses known initial poses rather than the more costly unknown-pose spectral matching [20].

Algorithm 1 details the complete MR-SLAM pipeline from sensor simulation through SLAM to dashboard update. A critical constraint is that Unity publishes only the `odom`→`base_footprint` subtree while SLAM Toolbox publishes `map`→`odom` (line 5). Allowing both to publish overlapping transforms caused TF tree conflicts and SLAM divergence. Fig. 4 illustrates the TF frame hierarchy and ROS 2 communication topology, showing this partitioning and the topic-level data flow.

A further integration challenge arose from a Quality of Service (QoS) mismatch between `ros_tcp_connector`'s RELIABLE clock and SLAM Toolbox's BEST_EFFORT subscription, which caused silent clock drops. A 5.0 s clock offset applied in the Unity `WallClockTime` component compensates for the accumulated drift between simulation and wall-clock time.

A custom `slam_stats_publisher.py` node queries each robot's map topic and publishes cell counts (free, occupied, unknown), map dimensions, and resolution as `Float32MultiArray` messages. The Unity `SLAMTelemetry` component then derives coverage area and scan rate locally from these messages to update the dashboard.

Algorithm 1 MR-SLAM End-to-End Pipeline

Require: N robots in Unity, SLAM Toolbox $\times N$, map_merge node
Ensure: Merged occupancy grid, live dashboard metrics
1: **// Sensor and TF Bridge (Unity $\rightarrow$ ROS 2)**
2: **for** each robot $r \in \{1, \ldots, N\}$ **do**
3: Cast 180 rays over 90° arc at 10 Hz $\rightarrow$ `/robot`$_r$`/scan`
4: Publish $T_{r/\text{odom} \rightarrow r/\text{base\_footprint}}$ at 20 Hz
5: **Skip** $T_{\text{map} \rightarrow r/\text{odom}}$ {SLAM owns this}
6: **end for**
7: Publish `/clock` at 100 Hz with wall-clock offset δ_t
8: **// Per-Robot SLAM (ROS 2)**
9: **for** each robot r **do**
10: SLAM Toolbox$_r$ $\leftarrow$ subscribe `/robot`$_r$`/scan`
11: SLAM Toolbox$_r$ $\rightarrow$ publish `/robot`$_r$`/map`, $T_{\text{map} \rightarrow r/\text{odom}}$
12: **end for**
13: **// Map Fusion and Telemetry (ROS 2)**
14: map_merge $\leftarrow$ {/robot$_1$/map, …, /robot$_N$/map}
15: map_merge $\rightarrow$ `/merged_map` using known initial poses
16: **for** each robot r **do**
17: Classify cells: free (v=0), occupied (v>50), unknown (v=−1)
18: Publish $[n_f, n_o, n_u, n_t, w, h, res]$ to `/robot`$_r$`/slam_stats`
19: **end for**
20: **// Dashboard Update (Unity)**
21: **for** each robot r **do**
22: coverage$_r$ $\leftarrow (n_f + n_o) \times res^2$
23: scan_rate$_r$ $\leftarrow$ LaserScanSensor publish rate
24: **end for**
25: **// Operator Teleop (Unity $\leftarrow$ Meta Quest 3 controller)**
26: Read joystick $\rightarrow$ apply deadzone $\rightarrow$ clamp (v, ω)
27: Publish `Twist` to selected robot's `/cmd_vel`

IV. Experiments

A. Setup

Experiments were conducted in a 6×8 m laboratory. The Meta Quest 3 scanned the room with MRUK to generate a spatial mesh for passthrough occlusion. Three simulated TurtleBot3 robots were placed at distinct starting positions. The pipeline was evaluated across five 9-minute sessions, with a single operator teleoperating all three robots sequentially in each. The operator was a graduate researcher with prior ROS 2 and Unity experience but no prior exposure to the MR-SLAM interface. Table III lists the configuration. An additional session in the same laboratory, recorded with the same hardware and software, provides the illustrative maps and timing distributions used in the figures of §IV-C.

TABLE III
Hardware and Software Configuration

Component	Specification
MR Headset	Meta Quest 3 (Snapdragon XR2 Gen 2)
Unity Version	2022.3 LTS (Android)
ROS 2	Humble Hawksbill (Ubuntu 22.04)
SLAM	SLAM Toolbox [18] (async) ×3
Map Merge	`multirobot_map_merge` (known poses)
Robot Model	TurtleBot3 Burger ×3 (simulated)
LiDAR	180 rays, 90° FoV, 10 Hz
Bridge	`ros_tcp_connector` (TCP:10000)

B. Metrics

Five metrics were recorded using rosbag2 and `slam_stats_publisher`, selected to assess spatial fidelity (coverage, overlap agreement), communication reliability (scan rate), and timing consistency (TF jitter). **Coverage**: fraction of grid cells classified as free or occupied;

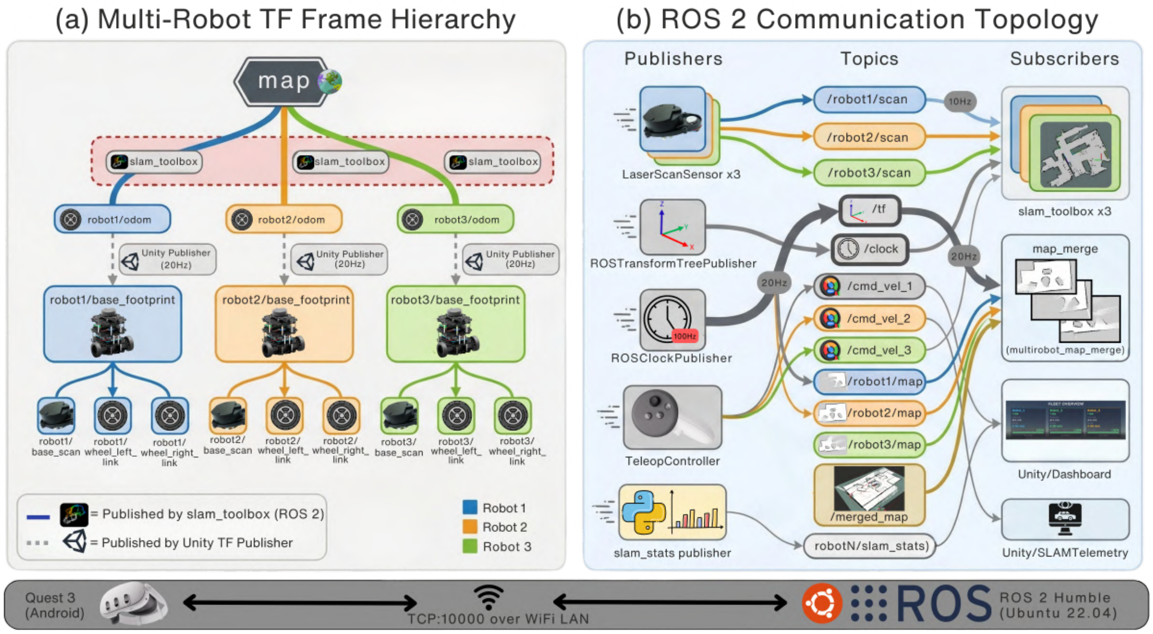

Fig. 4. (a) TF frame hierarchy: solid arrows = SLAM-published (map→odom), dashed = Unity-published. Red callout marks the partitioning constraint. (b) ROS 2 topic graph with publishers, subscribers, and data rates.

coverage area is the observed-cell count multiplied by the squared cell resolution. **Cross-instance occupancy consistency**: for each robot pair, the percentage of spatially overlapping cells with identical occupancy values. With known initial poses and a shared `map` frame, this characterizes whether the three independent SLAM Toolbox solutions agree in shared regions, not whether unknown-pose merging would succeed. **Scan rate**: arrival frequency of `LaserScan` messages at SLAM Toolbox. **TF jitter**: absolute deviation of each Unity-published transform's clock difference from its per-link mean (excluding SLAM-published `map`→`odom` links). **Teleop profile**: linear and angular velocities issued by the operator.

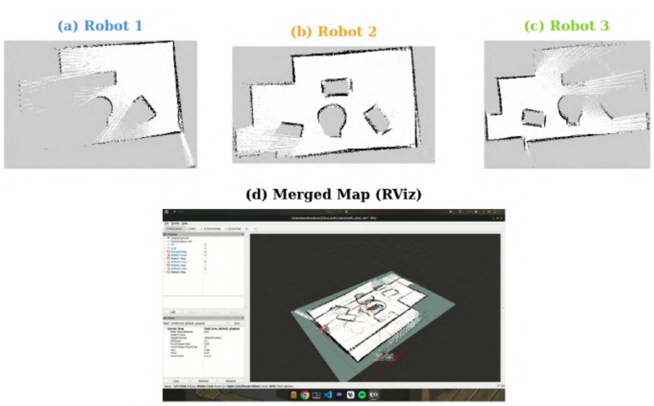

Fig. 5. Per-robot occupancy grids (top) and merged map from RViz (bottom). White: free, black: occupied, gray: unknown.

C. Results

Across the five primary sessions, merged-map coverage averaged $17.9 \pm 0.8\,\mathrm{m}^2$, with absolute coverage in m^2 depending on operator path and the specific room layout. Fig. 6 shows cumulative coverage over time in the additional session, in which the merged map reached $26.7\,\mathrm{m}^2$ ($\sim$65% of a $41\,\mathrm{m}^2$ grid). Robot 1 was driven first through the central area, followed by Robot 2 (left) and Robot 3 (right); coverage growth slowed as robots revisited previously mapped regions.

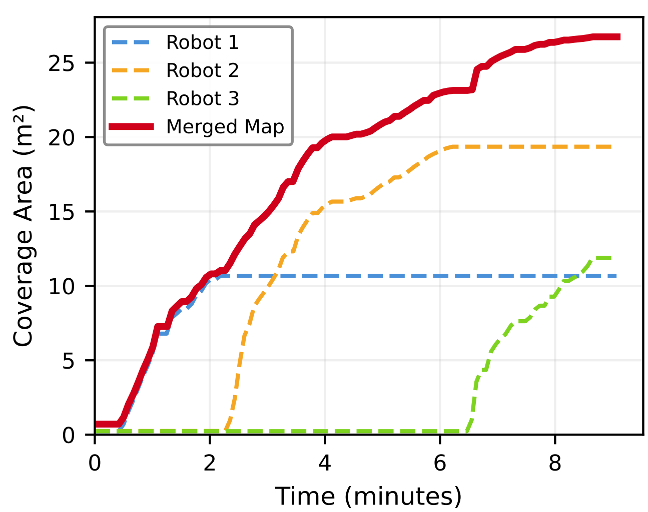

Fig. 6. Cumulative map coverage over time for each robot and the merged map.

Scan delivery averaged $8.83 \pm 0.16\,\mathrm{Hz}$ across the five primary sessions. Fig. 7(a) shows the scan-rate trace from the additional session, with a mean of 9.06 Hz and a minimum of 8.36 Hz when the operator switched between robots, within 10% of the commanded 10 Hz. TF jitter (Fig. 7b), measured in the additional session, had a median of 6.3 ms (mean 6.8 ms, P95: 16.3 ms, max: 26.2 ms) over 972 samples from 18 Unity-published links, consistent with the 82 ms round-trip reported by Quest2ROS [37] and well below latency thresholds that cause SLAM divergence.

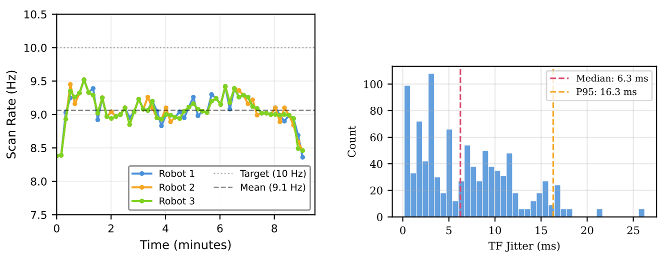

Fig. 7. (a) Scan reception rates at SLAM Toolbox for each robot. (b) TF jitter distribution; median 6.3 ms, P95: 16.3 ms.

Cross-instance occupancy consistency between robot pairs averaged $94.7 \pm 0.5\%$ across the five primary sessions. In the additional session this value was 95.4% (minimum 92.4%) across 35 measurements where spatial overlap existed. Because the merge uses known initial poses and a shared `map` frame, this metric measures whether the three independent SLAM Toolbox instances produced coherent grids in shared regions, not the quality of the merge operation; it serves as a consistency check that no single instance has diverged. Fig. 5 shows the per-robot occupancy grids and the merged map from the additional session. Table IV summarizes the primary-session statistics.

TABLE IV
QUANTITATIVE RESULTS ACROSS FIVE SESSIONS

Metric	**Value (mean $\pm$ std, n=5)**
Merged coverage	$17.9 \pm 0.8\,\mathrm{m}^2$
Cross-instance consistency	$94.7 \pm 0.5\%$
Scan rate	8.83 ± 0.16 Hz
TF jitter (additional session)	6.3 / 16.3 ms (median / P95)
Number of robots	3
Session duration	$\sim$9 min each

Fig. 2 shows the operator's passthrough view: virtual robots appear on the laboratory floor with correct perspective and occlusion. The dashboard panels (Fig. 3) allow monitoring mapping progress without removing the headset.

V. Discussion

Centralized map merging. Initial development attempted decentralized SLAM following Swarm-SLAM [8], but the geometrically clean raycasted scans lacked the feature distinctiveness needed for reliable cross-robot place recognition. Independent SLAM Toolbox [18] instances with centralized map merging proved more practical, at the cost of requiring known initial poses. This is acceptable in structured environments such as warehouses or offices; unstructured settings would require unknown-pose merging [20] or distributed optimization [24].

Scalability and deployment. The system supports three robots, limited by the operator's sequential teleoperation rather than computation (each SLAM instance is independent). Nav2 integration would decouple fleet size from operator bandwidth [47], [48]. Intended deployment scenarios are co-located fleet supervision tasks such as building-wing inspection and warehouse-aisle monitoring, where the operator is physically present in the workspace. We do not claim suitability for remote beyond-line-of-sight operation, where a non-passthrough rendering would be more appropriate. The effect of MR on operator cognitive load compared to conventional 2D displays is left to the planned NASA-TLX study (§VI).

Limitations. The simulated LiDAR raycasts against Unity colliders; physical deployment would introduce sensor noise, odometry drift, and dynamic obstacles. MRUK's spatial occlusion uses an on-device mesh of the operator's room, reconstructed by the headset and independent of the robots' SLAM map. Headline numbers come from five 9-minute sessions with one operator in one laboratory; the figures use an additional session at the same location. Multi-operator and cross-environment evaluation remain future work.

Spatial interaction. The dashboard uses MR primarily through world-space anchoring of per-robot panels (Fig. 3); the panel content itself is conventional 2D UI. Future MR-native extensions include gaze- or proximity-driven panel selection, in-situ rendering of the occupancy grid aligned to the operator's physical floor through passthrough, and hand-gesture waypoint placement in world space.

VI. Conclusion and Future Work

This paper presented MR-SLAM, a mixed reality system for multi-robot SLAM in which an operator wearing a Meta Quest 3 teleoperates three simulated TurtleBot3 robots and monitors mapping progress through spatially anchored dashboards with passthrough occlusion. The system bridges Unity to ROS 2 Humble via `ros_tcp_connector`, running three independent SLAM Toolbox instances with real-time map merging. Across five 9-minute sessions, the system delivered scans at 8.83 ± 0.16 Hz, mapped $17.9 \pm 0.8\,\mathrm{m}^2$ of merged occupancy, and reached $94.7 \pm 0.5\%$ cross-instance occupancy consistency; an additional session contributed the 6.3 ms median TF jitter measurement. The end-to-end pipeline (Algorithm 1), including TF partitioning and QoS-aware clock synchronization, serves as a reusable reference for Unity–ROS 2 SLAM integrations.

Future work includes a multi-operator NASA-TLX study with a monitor-based baseline, unknown-pose map merging via spectral matching [20], deployment on physical TurtleBot3 hardware, potential industrial validations in relevant industrial contexts [49], and MR-native interaction features with Nav2 integration [47], [48].

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